Passive Precision Time over BLE V14 Candidate Service and Profile Proposal

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Abstract

Passive Precision Time over BLE (PToB) proposes a common clock service for heterogeneous Bluetooth Low Energy listeners. Complete periodic advertisements identify physical timing events; subsequent corrections let receivers discipline their own clocks without a transaction for every timestamp. Grandmasters use authenticated broadcast discovery, independent source qualification and a designated HOT successor. Relay clocks regenerate timing through causal local maps while preserving root provenance. Four authenticated coarse metadata bytes convey the serving source's time zone and effective civil-time offset, without changing precision timestamps. This proposal defines the service, GM7 succession, timing and trust boundaries, wire-profile selection and conformance work needed for interoperable adoption. Precision, failover availability and energy performance require separate implementation and measurement evidence.

1 Purpose, scope and roles

The service lets an endpoint identify a time source, maintain a local clock between observations, report provenance and withdraw unavailable time. Receivers may range from coarse callback consumers to hardware timestamping instruments. Packet resolution alone makes no accuracy promise.

This is a proposal for discussion, not an adopted specification, SIG submission or qualified product. MUST, MUST NOT and SHOULD express candidate requirements. The reference profile is GM7v4+S2+C1+TZ1 with G1/GR3 root timing and RC7/profile5 relay timing. Versioned contracts and public reference codecs define the detailed byte bindings [1, 4, 3]. Implementation support and physical qualification remain distinct.

Role	Function and authority
ACTIVE GM	Independently qualified source currently originating mesh time.
HOT GM	Designated successor with qualified own source and a usable same-scope shadow clock; conditionally prepared for takeover.
COLD / following GM	Receives mesh time and maintains its own source; has no fast-takeover entitlement.
Relay clock (RC)	Follows one authenticated parent and regenerates downstream timing; receives no GM election or HOT authority.
Client / leaf	Consumes an admitted source. Capture capability determines its timing tier; a parent constraint remains binding.

A device's capabilities, selected profile, current role, functional clock availability and precision quality MUST remain separately observable. A fitted clock or display label cannot create independent-source eligibility. A role or source transition fences affected authority, captures and maps before replacement service.

2 GM7 election, following and failover

GM control uses authenticated connectionless ANNOUNCE (ANN), STATUS and WITHDRAW records. ANN names the ACTIVE source and one preferred successor. Each following GM retains its own GNSS/PPS source, listens to the selected GM's timing and reports independent eligibility and actual shadow preparation in ordinary STATUS. A qualified nominee echoes HOT in its next STATUS. These ordinary broadcasts carry GM coordination and nomination [1].

Eligibility precedes ranking. The reference source requires 60 seconds of continuous independent GNSS/PPS qualification. Ordinary initial election then requires 30 seconds of quiet observation plus deterministic backoff up to 500 ms: a healthy cold start normally takes about 90 seconds plus backoff. A GM lacking its own qualified source may follow time but MUST NOT originate it. Rank eligible candidates lexicographically by priority1, clock class, accuracy, variance, priority2 and stable identity, with boot identity resolving incarnation ties. Preference alone is never HOT confirmation.

HOT readiness requires a current same-scope delivered shadow map and fit, live correction/source evidence, qualified own GNSS/PPS and a physical phase enclosure within the 2 ms scheduling policy. Strict precision LOCK is a separate assessment. Nomination, diagnostics and displayed RMS cannot substitute for those readiness predicates.

Transition	Required behavior
Selected GM silence	Only the designated, still-qualified HOT uses the fast path: trigger at 2.5 s after last advancing selected-GM communication; target first completed G1 within 3 s of that endpoint. Actual capture, drain, source and reservation gates still apply.
Explicit YIELD	Stop-before-start at the immutable boundary, 750 ms after the original source sample; translate it through the retained physical map. Three copies 250 ms apart cannot move it.
WITHDRAW / source fault	Stop positive output immediately; attempt three immutable withdrawal copies within their original 1 s negative lifetime. A successor still needs its own authority and source readiness.
No usable HOT	An independently qualified COLD GM uses ordinary 30 s quiet election plus backoff. Held remote time cannot create source eligibility.
Fresh selected contact	Advancing ANN, authenticated self-ACTIVE ROLE or direct timing header inhibits silence-only takeover. Duplicate/reordered traffic does not renew liveness.

GM following continues across control loss. Actual authenticated timing scope, physically matched corrections and their original expiry bound delivered time; a remembered listening identity grants no map. Map loss alone is not transmitter failure. The 3 s objective requires physical qualification.

Mesh lineage and convergence

At first valid serving reservation, sample full 64-bit functional UTC as mesh birth and record the founder. Commit ACTIVE only after successful bounded ANN transmission; track actual RF completion separately. A possibly exposed claim retains its lineage even if completion fails. Successors inherit birth/founder and advance the leader term; boot, source epoch, capture generation and train remain separate scopes [2].

When ACTIVE meshes meet, earlier birth wins; exact-birth ties use founder, then term/identity/boot ordering. The newer mesh stops positive output before following. The older ACTIVE keeps its service and nominee. Disconnected partitions can serve separate independently eligible sources; convergence does not promise consensus across absent links. Birth is a functional ordering value, not proof of calibrated UTC.

3 Periodic timing and relay regeneration

Timing uses LE periodic advertising, with complete application records at a nominal 62.5 ms interval (16 Hz). Acquisition uses extended-advertising discovery; synchronization is distinct from connecting. An application record may span controller reports or link-layer fragments, so adapters MUST deliver bounded whole records and preserve their event identity.

An original publication identifies a physical transmission without predicting its own future TX timestamp. A later record carries zero, one or two corrections naming earlier publications, periodic events and physical data channels (0–36). The receiver matches a correction to the saved physical RX capture of that exact original. Callback arrival, follow-up RX and worker processing time are different events and MUST NOT replace that capture.

The association scope includes authenticated group, root identity/boot/source epoch, immediate sender identity/boot, train incarnation, capture epoch, publication, periodic event, channel and adapter/PHY context. Missing originals, ambiguous copies, replay, nonpreceding corrections and retired scope are refused. Counter wrap is valid only within its bounded live scope; overflow cannot alias an earlier event.

Let $M_g(x)$ map local counter x to source time in immutable generation g. A completed pair supplies the anchor

$$M_g(x_{\mathrm{RX}}) = T_{\mathrm{TX}} + d,$$

where d is the declared path/reference treatment with its uncertainty. A new map becomes usable only after correction processing completes. Earlier reads or witness events retain the map then available; later evidence cannot improve past output.

Boundary clocks and paths

An RC disciplines its own counter and originates a downstream train. Each correction uses the map effective at its own TX event and preserves that generation. It MUST NOT present the parent's physical event as its own. Root identity, boot, source epoch, original issuance/expiry, UTC mapping and uncertainty remain provenance; sender boot, train and capture context describe the new transmitter.

RC7 authenticates a dynamic bounded path of root and at most two relay transmitters, including identities, boots, trains, capture epochs and slots. It rejects loops, incompatible scope and invalid parent/depth relationships. A leaf does not acquire relay or GM authority through admission. Root position zero and nominal RC positions at 15.625, 31.250 and 46.875 ms are scheduling policy, not PAwR response slots or proof of actual controller placement. Missed preparation advances by whole frames. Port-specific TX estimates require an explicit timing tier and uncertainty [3].

Validity and continuity

Functional availability, local discipline age, inherited age, independent source permission, admission/demand and precision quality are separate quantities. RC7 renews local context around 300 s with at most a 360 s lease; these durations do not make timing fresh for minutes. The root source envelope is bounded to at most 1.65 s. Each consumer also applies its actual correction, fitter and profile expiry limits.

Holdover needs a declared oscillator model, elapsed age and supported error budget. It MUST NOT renew root permission or GM entitlement. Unknown capture, path, reference or rate uncertainty propagates as UNKNOWN. Reboot, source/counter epoch change, revoked permission or arithmetic fault fences affected output. Within a continuous valid scope, delivered time is monotonic; reacquisition discontinuities are explicit. TZ presence has its own metadata lease and cannot extend any timing lifetime.

4 Wire profile and authentication

The selected profile uses the following explicit discriminators and sizes. Counts are complete application records; GM body sizes exclude its four-byte manufacturer AD wrapper. The development manufacturer value is `0xffff`; production identifiers and interoperable service/bearer assignments require SIG-facing work. Detailed layouts, codecs and public synthetic fixtures are cited rather than duplicated here [3, 4].

Record	Selected revision / bytes	Purpose
Root timing	GR3, version 3; 190-byte AD	Root envelope, physical originals and corrections; 16 MHz wire units.
Relay timing	RC7, version 7/profile 5; 190-byte AD	Regenerated TX nanoseconds, map generations, cumulative age and authenticated path.
GM control	G7V4, revision 4; 182-byte body / 186-byte AD	ANN, STATUS, WITHDRAW; complete lineage, nominee and four-byte TZ.
GM observations	G7V4, revision 4; ROLE 168 / GNSS 156 body bytes	Bounded role/source diagnostics; neither record grants timing or HOT authority.
Direct client admission	JOIN revision 5; 168 bytes	Authenticated membership, coarse bootstrap/renewal and TZ.
Relay admission	Coarse revision 8; 216 bytes	Local request/response, path and bounded demand/source context, TZ.
Relay capability	Coarse revision 8; 148-byte AD	Discovery, existing admission challenge and current TZ.

GM control uses big-endian integers in its existing fields; TZ is explicitly little-endian. G1/GR3, RC7 and JOIN fields use their declared little-endian layouts. Signed values use two's complement. Do not serialize native structures or infer a decoder from length alone. Exact framing, version/profile, kind, reserved zeros and group epoch are mandatory. A 16 MHz unit is 62.5 ns representation, not a physical oscillator or accuracy requirement; RC nanoseconds likewise specify representation.

The G7V4 body begins with magic (bytes 0–3), revision (4), kind (5), exact body length (6–7), flags (8–9), reserved zero (10–11), group ID (12–19) and key epoch (20–23). ANN/STATUS/WITHDRAW retain the full source/lineage fields; timing epoch is at 158–161, TZ at 162–165 and final tag at 166–181. ROLE/GNSS carry no TZ extension.

All selected coarse records use a provisioned 32-byte group PSK and HMAC-SHA256 truncated to its first 16 bytes. For eight-byte ASCII domain D, group context C and complete pre-tag body B,

$$\text{tag} = \text{first}_{16}\big(\text{HMAC–SHA256}(K, D\|C\|B)\big),$$

where C is context-version byte 1, u64 little-endian group ID and u32 little-endian key epoch. Domains are `PTOBTZ04` (GM), `PTOBTZJ5` (JOIN), `PTOBTZR8` (relay control) and `PTOBTZC8` (capability). Precision keeps its own packet/origin/path authentication; a valid coarse tag cannot authorize a precision sample. GR3 origin coverage is start 14, length 60, i.e. bytes [14:74].

A receiver MUST select the complete profile explicitly and reject incompatible coarse revisions. Equal packet size or valid precision bytes do not admit a mismatched coarse source. Version changes need explicit discriminators, domains and new vectors; reserved bits cannot silently acquire meaning.

5 Admission, receiver behavior and trust

Timing broadcasts require no per-timestamp transaction. Receivers MUST discover local meshes, separate trusted credentials from runtime mesh selection, and authenticate admission afresh after context changes. The shared PSK establishes membership/integrity; every holder can originate statements. Public decoding alone provides neither trusted discipline, individual attestation, secrecy nor keyless verification.

Ordinary clients use direct JOIN or relay-local admission; GMs follow broadcast-selected timing. JOIN binds identities, boots, fresh nonces and forward sequences in a bounded exchange. Coarse extraction retains source uncertainty and subtracts the entire causal RTT (at most 250 ms). Its replay anchor MUST NOT train the precision estimator. Relay admission binds path, parent boot, challenge, nonce, publication floor and bounded source context [3].

Demand and output availability are independent: an authenticated request can start acquisition while timing is unavailable. A not-ready response grants no timing. Renewing same-scope context preserves still-valid captures/maps when resources allow but MUST NOT reset source age or resurrect expired output. Ordinary source advertisements and current usable source evidence, not a long demand lease, determine whether an ACTIVE GM can admit clients.

Receive procedure

1. Reassemble a bounded whole record with transport and capture identity. Check exact length, discriminator, version/profile, kind and canonical reserved fields.
2. Verify group/key context and all applicable packet, origin and path tags. Bind the record to the selected sender, boot and admitted source/path scope before committing state.
3. Apply replay, sequence/publication floors, terminal/term fences and the original receipt/source deadlines. Reject foreign, duplicate, conflicting, expired or incompatible scope without renewing state.
4. Save a valid original's physical RX capture. Accept a later correction only for an unambiguous preceding original in that scope; check timestamp conversion and arithmetic bounds.
5. Publish a new immutable map only after processing completes. Report functional availability, source provenance, cumulative age and known or unknown uncertainty separately. Preserve refusals and unavailable opportunities.

For ordinary GM7 broadcasts, original receipt time bounds local observation lifetime; polling, duplicate copies and delayed processing cannot extend it. Authentication and retained floors cannot prove the age of a previously unseen recording or advancing recorded stream. This first-contact replay limitation is explicit, including after eviction from bounded peer memory. Durable replay across arbitrary state loss, key rotation/revocation, compromise recovery, asymmetric public verification and delay-attack resistance need production security-profile work.

Clock maps, management observations and radio ownership have different lifetimes. Capture callbacks use bounded work and storage; queue overflow must be observable and cannot fabricate captures or precision. Supported controller reservations, actual drain and no-double-output fences govern serving transitions. Diagnostic collection ending MUST NOT stop healthy service. Explicit stop, fault containment and update drain remain bounded operations.

The time scale is a POSIX UTC estimate with an authenticated source-coordinate mapping. Leap/TAI handling and any wired PTP/gPTP gateway need an explicit separate scale, delay and uncertainty contract. Local civil time never replaces UTC event identity.

6 Time zones and daylight-saving support

The serving GM publishes one atomic pair in authenticated coarse metadata:

Bytes	Field	Meaning
0–1	`tz_id`, u16 LE	Stable append-only PToB identifier mapped to an IANA zone name.
2–3	`tz_offset_min`, i16 LE	Current effective whole minutes east of UTC; valid range -840 to +840.

ID 0 is unknown, ID 1 is UTC (offset zero), and ID 2 is America/Chicago. The canonical unknown pair is `00 00 00 80`, representing (0, -32768). Chicago CDT is `02 00 d4 fe`, or (2, -300); CST is (2, -360). IDs identify regions, not seasonal abbreviations, and are PToB assignments rather than IANA numeric identifiers [4, 5].

Receivers format local civil time as UTC plus offset×60 seconds, with checked signed arithmetic. They use the transmitted effective offset and need no zone database or DST calculation. An unfamiliar nonzero ID with a valid offset remains usable and forwards unchanged. Invalid metadata normalizes to unknown without invalidating otherwise valid authenticated UTC; malformed framing or authentication failure rejects the packet. Unknown/expired metadata falls back to UTC.

Selection, ordering and propagation

The GM may use automatic location-based selection, a manually selected zone/calendar, or an explicit fixed offset. The reference automatic selector uses its own fresh GNSS fixes, conservative region coverage and a finite calendar validity window. Ambiguous/stale location permits only the declared bounded hold; an expired calendar produces unknown immediately. Standby GMs may prepare their own zone; on takeover the new ACTIVE publishes its own selected pair, including unknown if unavailable. Geographic coverage and calendar support are implementation capabilities, not universal protocol guarantees [6].

An RC learns the pair only from its selected authenticated parent and forwards it unchanged in coarse capability/admission records. Direct updates require ordered selected-GM ANN with current root/boot/epoch scope and replay checks. Initial JOIN metadata cannot overwrite a later accepted ANN. Relay changes use the existing monotonic admission challenge: a higher token can change the pair; an equal token renews only an identical pair. Older or equal-token conflicting updates are refused. No TZ counter or transition schedule is added.

TZ has a local presence lease of three nominal parent advertising periods: 750 ms for a 250 ms GM cadence, 300 ms for a 100 ms RC cadence. Matching authenticated parent advertisements may renew that lease independently of precision corrections, lock and map expiry. Each hop renews its own presence lease. Polling cannot renew it. After expiry the live value is unknown; the last accepted pair may support matching-presence renewal. An explicit unknown update clears that pair, and parent/path replacement fences the old context.

Receivers must continue receiving coarse updates after timing synchronization. Precision presence can renew a cached pair but carries no changed pair. A DST transition changes the effective offset on fresh coarse reception; simultaneous scheduled display changes are not promised. Repeated local hours retain distinct UTC event identities.

TZ MUST NOT alter precision G1/GR3/RC7 payloads, UTC mappings, clock discipline, PPS, source/mesh epochs, replay comparisons, USB UTC or chrony inputs. The precision field `utc_offset_ns` maps source coordinates to UTC and MUST NOT be repurposed for civil-time offset. TZ grants no timing permission. Other bearers require their own explicit metadata contract.

7 Conformance targets and proposed SIG work

Conformance is layered: byte interoperability, state/authority correctness, radio-event association, functional availability and independently measured precision require separate results. No codec, model, native build, fitted residual or successful radio request establishes all of them.

Area	Required verification
Wire and bearers	Exact versions/lengths/tags, canonical padding, complete reassembly, malformed/refused vectors, transport/discovery binding and cross-vendor decoding.
GM7 authority	Independent qualification; initial election; actual HOT confirmation; source/control loss; immutable YIELD/WITHDRAW; physical stop-before-start; conditional 3 s first-G1 target; no-HOT fallback; partition merge.
Timing and relays	Unique TX/RX originals and corrections; capture/scope transitions, replay and wrap; causal map activation; original expiry; one- and two-relay paths; actual controller placement and loss recovery.
TZ and DST	Signed/fractional-hour offsets, unknown/future IDs, ordered direct/relay updates, withdrawal and exact lease expiry, parent/GM replacement, repeated/skipped local hours and unchanged UTC/precision bytes.
Precision and resources	Named output/reference points, calibration and uncertainty, bias and total RMS, tails, unavailable/missing counts, drift/holdover, coexistence, energy and sustained operation.
Security and lifecycle	Current admission, key/epoch/boot changes, bounded inventories, compromise/revocation and replay limits, diagnostic independence, explicit stop and fault/resource closure.

Reference implementations and offline tests support review of these mechanisms. Physical GM failover, sustained relay availability, cross-vendor interoperability, constrained listening and calibrated UTC/output bounds remain open qualification work. Report every population with its denominator, invalid/missing observations and failures. Internal clock agreement, fractional PPS phase and independently measured output remain distinct metrics. Unknown uncertainty stays unknown.

The proposed SIG work item is a passive clock-service/profile family with explicit event association, causal publication, authenticated authority separation and interoperable quality reporting. The production profile must settle assigned identifiers, discoverable service/capability definitions, connected-record fragmentation/MTU behavior, controller timestamp/scheduling support and security lifecycle. Claimed listening or energy tiers need measured acquisition, recovery and steady-service budgets.

Boundary-clock regeneration describes the architecture; IEEE PTP/gPTP media, BMCA, TSN, wired gateway behavior and Bluetooth product qualification require their own applicable specifications and tests. This discussion draft supplies concrete protocol behavior for review without claiming standards adoption or a universal accuracy tier.

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References

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[2] J. Yantis, *GM7 mesh-birth contract C1*. `protocol/gm7_birth_c1/README.md`. Logical initiation, possible RF exposure, successful completion and inherited lineage.

[3] J. Yantis, *Public timing and admission reference codecs*. `firmware/t096/protocol/`: `group_precision.c`, `relay_codec.c`, `rc_dynamic.h`, `rc_dynamic.c`, `group_join.h`, `group_join.c`, `gm7_codec.h`, `gm7_codec.c` and `group_auth.c`. Exact profiles and synthetic test fixtures, not hardware acceptance.

[4] J. Yantis, *Coarse time zone extension v1*. `protocol/time-zone-v1/README.md` and `vectors.json`; `docs/time-zones.md`. Four-byte metadata, selected coarse revisions, authentication and dynamic lease/ordering policy.

[5] J. Yantis, *PToB time zone registry and generator*. `tools/time_zones/README.md`; append-only IDs in `tools/time_zones/zones.json`. Input versions, licenses and calendar/geography limits accompany the generator.

[6] J. Yantis, *T096 GM automatic time zones*. `docs/t096/auto-time-zone-plan.md`. Current selector/calendar modes, receive lifecycle and separately scoped validation results. Receiver ports: `docs/esp32-time-zones.md`.